Reinforcement Learning
Lecture IX: Modern RL

Asad Jeewa
University of Cape Town



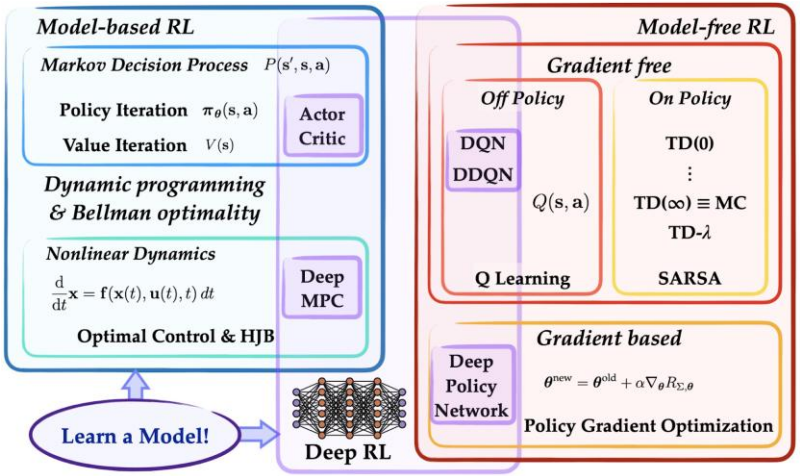
What we have covered

- RL Fundamentals
- MDPs, Value Functions
- Value and Policy Iteration
- Monte Carlo, Temporal Difference
- Q-Learning (tabular)
- Function Approximation (DQN)

What we did not cover

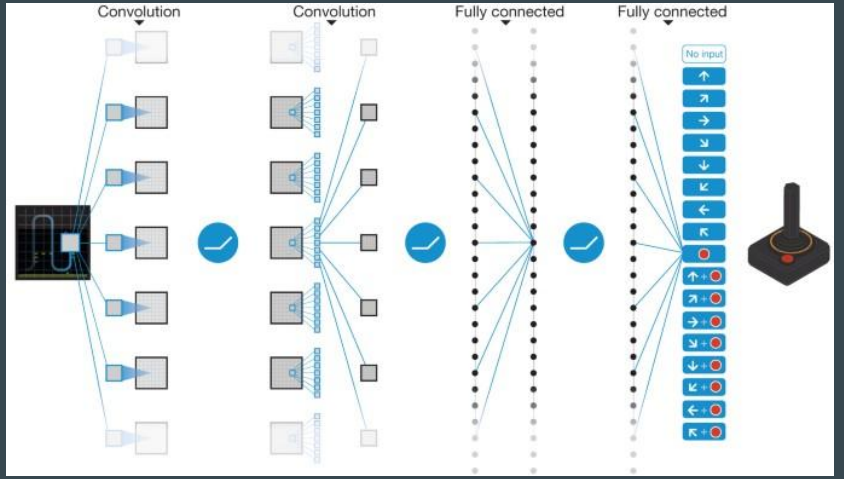
- Beyond discrete GridWorld
- Continuous Control
- True model-based RL
- n-step Bootstrapping
- Advanced DQN Variants
- Policy Gradient Methods
- Actor-Critic Methods
- Multi-Agent Reinforcement Learning (MARL)
- Multi-Objective Reinforcement Learning (MORL)
- Offline RL
- Etc.

The Bigger Picture



Beyond discrete GridWorld

- States: continuous or images
  - Images require feature extraction (CNNs)
- Actions: large discrete or continuous
- Rewards: may be sparse
- Dynamics: stochastic or delayed effects
- Observations: partial or noisy
- Horizon: short or long-term dependencies
- Exploration: safe vs risky regions



Continuous Control

- Actions are continuous (not discrete)
- DQN not suitable
- Methods:
  - DDPG (Deep Deterministic Policy Gradient)
  - TD3 (improved stability)
  - SAC (Soft Actor-Critic)
- Applications: robotics, control systems



True Model-Based RL

- We have looked at planning on a known MDP
  - Value Iteration
  - Policy Iteration
  - Classic planning assumes you already have T and R. You never query the environment
- True MBRL requires learning environment dynamics instead of only values/policy
- Estimate transition function $T(s, a, s')$ and reward function $R(s, a)$ from data
- Build an internal model of the environment
- Use the model to plan
  - Apply Value Iteration / Policy Iteration on learned T,R
- Example: Dyna

n-step Bootstrapping

- Combines TD and Monte Carlo: use n rewards before bootstrapping

$$G_t^{(n)} = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{n-1} R_{t+n} + \gamma^n V(S_{t+n})$$

- Update: $V(S_t) \leftarrow V(S_t) + \alpha[G_t^{(n)} - V(S_t)]$
- Trade-off: small n means more bias; large n means more variance
- Eligibility Traces keep a decaying memory of recently visited states so we can update all relevant past states at each step
  - Generalises n-step bootstrapping to online multi-step learning

Extending DQN

- DQN is a seminal algorithm that has been extended many times
- Double DQN: reduces overestimation
- Dueling DQN: separates value & advantage
- Prioritised Replay: samples important experiences more
- Rainbow DQN: combines multiple improvements

Policy Gradient Methods

- Learn policy directly: $\pi(a \mid s; \theta)$
- No need for max over actions
- Works well in continuous action spaces
- Example: REINFORCE
- Key Idea: optimise expected reward via gradient ascent

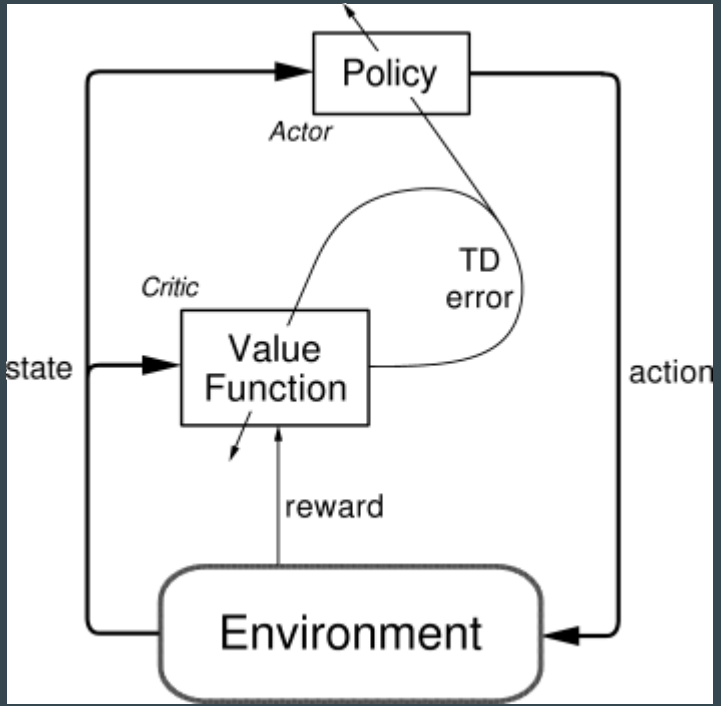
$$\max_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[R(\tau)]$$

Our goal is to find the weights θ **that maximize the expected return.**

https://huggingface.co/learn/deep-rl-course/en/unit4/introduction

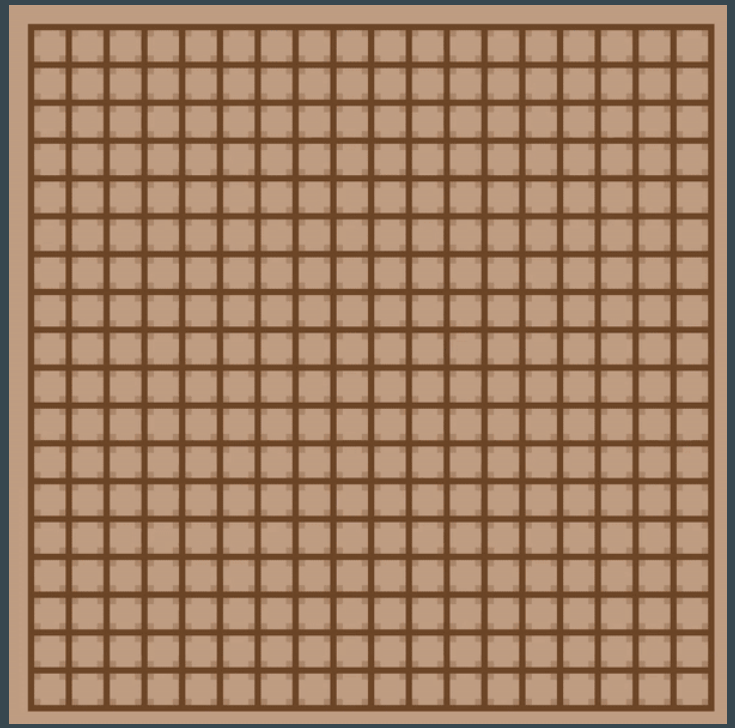
Actor Critic Methods

- Combine value-based + policy-based
- Actor: learns policy
- Critic: evaluates actions (value function)
- Benefits
  - Lower variance than policy gradients
  - More stable than pure DQN in some settings
- Examples: A2C, A3C



Multi-Agent RL

- Multiple agents learning simultaneously
- Environment becomes non-stationary
- Challenges:
  - Agents affect each other's learning
  - Credit assignment problem
  - Coordination vs competition
- Examples:
  - Self-driving traffic systems
  - Game AI (multi-player)

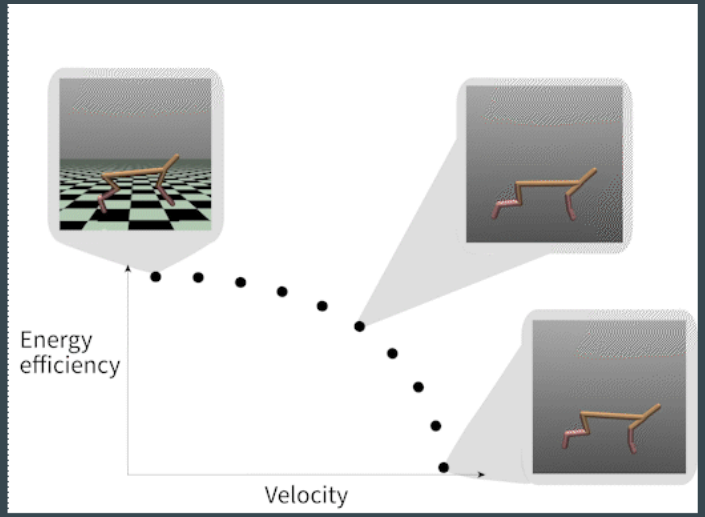


Cooperative vs Competitive MARL

- Cooperative: agents share a goal
- Competitive: agents have opposing goals
- Mixed: combination of both

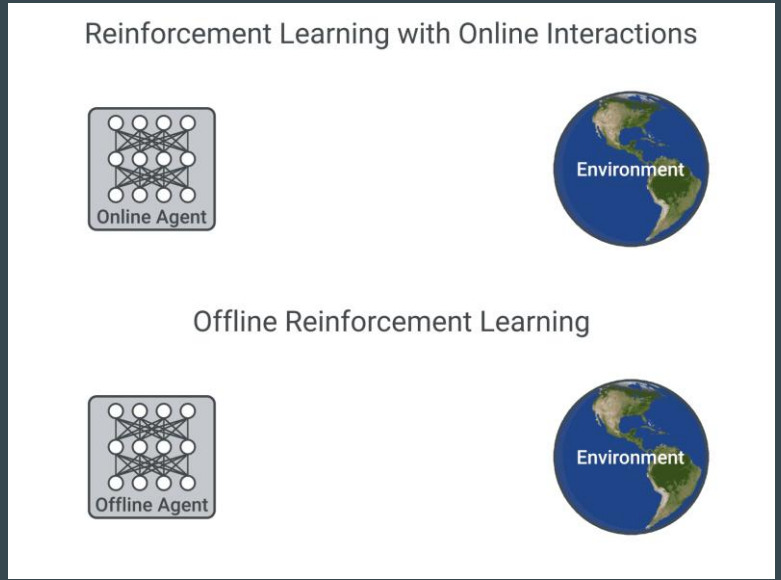
Multi-Objective RL

- Multiple objectives learned simultaneously
- Reward is a vector, not a scalar
- Challenges:
  - Conflicting objectives (trade-offs)
  - Choosing or learning preferences
  - Scaling to complex environments
- Examples:
- Autonomous driving (safety vs speed)
- Robotics (energy vs performance)
- Finance (risk vs return)



Offline RL

- Learn from fixed datasets (no interaction)
- Important when interaction is expensive/dangerous
- Challenges:
  - Distribution shift (data ≠ deployment environment)
  - Overestimation of unseen actions
  - Limited exploration (can't try new actions)
- Examples:
  - Healthcare
  - Finance
  - Autonomous driving



Where to Next?

- Try implementing more complex algorithms such as PPO or SAC
- Explore MARL/ MORL environments
- Look at how we can apply RL to a real-world problem
- Look at scientific papers (Google Scholar)

Practical RL Tools and Repos

Area	Algorithms	Environments/ Datasets
RL	Stable-Baselines3, CleanRL	OpenAI Gymnasium
MARL	Rllib (Also RL)	PettingZoo
MORL	MORL-Baselines	MO-Gymnasium
Offline	CORL	Minari

Additional Resources

- Sutton & Barto (1998), Reinforcement Learning: An Introduction,
- https://huggingface.co/learn/deep-rl-course/unit4/introduction
- https://spinningup.openai.com/en/latest/
- https://www.youtube.com/@DeepLearningIndaba
- https://rail.eecs.berkeley.edu/deeprlcourse/